Gautam Gupta

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Education

Master of Science (MS) in Management, George Mason University, VA (August 2017- June 2018)
Bachelor of Science (BS) in Healthcare Administration and Policy, George Mason University, VA (August 2013- May 2017)

Certifications

AWS Cloud Practitioner Certified, Amazon Web Services (AWS)

- Registration #: 475207070
- Certification ID: AWS04329983
- Validation #: a2fe9d03a0dd46f1804dea88fbbeb4f5
- Certified Badge Verification URL: https://www.credly.com/badges/ca6003c2-3436-47da-bc4c-1abf890fc1fa/public_url

Microsoft Azure Data Fundamentals Certified, Microsoft

- Demonstrated understanding of core data concepts and related Microsoft Azure services
  Certification #: 1478-3880
  Certification ID#: 992289801
  Certified Badge Verification URL: https://www.credly.com/badges/cbb09bda-b4b8-4d02-89c6-c04ddd88a9c3/linked_in?t=rl1xth

Information Technology Infrastructure Library (ITIL) 4 Foundations in IT Service Management Certified, Axelos

- Demonstrated understanding of IT service management fundamentals
  Certification ID#: GR671327256GG
  Candidate #: 9980088928169061

The Open Group Architecture Framework (TOGAF 9) Standard Certified, The Open Group

- TOGAF 9 Standard Certified Badge Verification URL: https://www.credly.com/badges/d37ad268-0a66-4cf6-a1b9-81bf143482c3

  - TOGAF 9 Certified
    - Demonstrated understanding of Enterprise Architecture (EA) and Architecture Development Method (ADM) application for practical use in real world scenarios (October 2021)

  - TOGAF 9 Foundation Certified
    - Demonstrated understanding of Enterprise Architecture (EA) and Architecture Development Method (ADM) foundations and theory (September 2021)

Data Analytics Certification, George Washington University

- Successfully completed a 6 month long, part time rigorous bootcamp while working full time, which covered a plethora of skills used in the domain of data analytics

Security Clearances

Security Clearance Level – Top Secret, Department of Defense (DoD) – Present/Active

Experience Summary

An experienced Enterprise Data Architect (in Information Technology) with technical and business architecture work involving:

- Enterprise Architecture Frameworks and Solution Architecture

        - Department of Defense Architectural Framework (DoDAF), The Open Group Architecture Framework (TOGAF), Scaled Agile Framework (SAFe), System Engineering Lifecycle (SELC), Acquisition Lifecycle Framework (ALF), Risk Management Framework (RMF), Development and Operations (Dev Ops), Medallion architecture (bronze/silver/gold layers)
    - Data Architecture/Modeling, Gap Analysis, Data Analytics, Data Warehousing and Cloud
        - Dimensional and Relational Data Modeling (star and snowflake schemas), Extracting/Transforming/Loading (ETL), Data Visualization, Data Cleansing, Data Profiling, Data Governance, Database Creation and Manipulation, Data Vault Modeling
        - Tools in Azure include SQL Database, Data Factory, Data Storages, Databricks, Data Lake Gen2
    - Agile/Ad Hoc and Waterfall Environments
    - Microsoft Suite
        - Excel, Power Point, Word, Outlook, Teams
    - Diagramming and Data Modeling Tools
        - System Architect (SA), Cameo, Lucid Charts, Innoslate, Erwin, EKR ARIS
    - Data Analytic/Visualization and Database Tools including:
        - Semantic Open-Source Software (SEMOSS), Tableau, H2 (relational SQL console), Power BI, PostgreSQL, Microsoft SQL Server Management Studio (SSMS), Army Vantage, AWS Redshift, and Collibra (for data management)
    - Coding Languages
        - Proficient with Structured Query Language (SQL), Familiar with Python
    - Model Based System Engineering (MBSE) Diagramming
        - Unified Modeling Language (UML), Lifecycle Modeling Language (LML)
        - Business Process Modeling Notation (BMPN)
    - IT Service Management (ITIL)
    - Large Language Model (LLM) tools of CamoGPT, NiprGPT, AskSage, and more to leverage Artificial Intelligence capabilities

<u>Clientele's</u>

- Department of Defense (DoD)
    - Army, Defense Health Agency (DHA)
- Department of Homeland Security (DHS)
    - Customs Border Protection (CBP)
- Veteran Affairs (VA)
    - Veteran Health Administration (VHA)
- Department of Commerce (DOC)
    - National Oceanic Atmospheric Administration (NOAA)

<u>Work Experience</u>

Enterprise Data Architect, ECS Tech (December 2023– Present)

Army Global Force Information Management (GFIM) Project/ Department of Defense (DoD)

- Developed DoDAF based conceptual (DIV-1), logical (DIV-2), and physical (DIV-3) data models using Erwin data modeling tool for Training and Readiness domain entities.
- Used Army Vantage big data platform to produce analytics and Collibra tool for data management.
- Conducted data mining and data call workshops with key Stakeholders.
- Developed the corresponding data reference model to support the reuse of data.
- Established the plan for migrating legacy data into a new data construct.
- Planned, designed, and managed the implementation of data structures, and relational database best practices.
- Provided recommendations on data migration best practices and conduct data source analysis.
- Briefed team on Microsoft Azure concepts in learning sessions (showed how SQL Server can connect to Azure SQL Database and be accessible with Azure Data Factory, Data Bricks, and how Blob features with Data Lake Gen2 can be used (Azure Data Factory for an ETL from Data Lake into Azure SQL Database, using data flows within Data Factory, Data Bricks for Data Warehousing capabilities, etc).

- Worked with EKR ARIS diagramming tool and Erwin to create Unified Modeling Language (UML) class diagrams for Training and Readiness domains (conceptual, logical, and physical data models).
- Collaborated with stakeholders, including software developers, system architects, and business analysts, to gather data requirements and translate them into Unified Modeling Language (UML) class diagrams.
- Used Large Language Model (LLM) tools engaging in research of Army laws, regulations and policies and extracting ammunition management relevant ones into Artificial Intelligence (AI) platforms such as CamoGPT and AskSage, along with establishing metrics for those LRP, as well as used LLMs to create models.

Enterprise Data Architect, Significance Inc. (May 2022– December 2023)

Control System Governance Office (CSGO) project, Department of Defense (DoD) / Army

- Worked in the enterprise architecture and data analytics division to create a data strategy for showcasing and storing high risk Facility Related Control System (FRCS) information for stakeholders.
- Created a FRCS Data Strategy Plan and Implementation Plan Gantt Chart, incorporating the data elements of data architecture, data cleansing, data conversion, data maintenance and governance model for a proof of concept.
- Followed the Risk Prioritization Framework (RPF) guidelines to categorize high risk control systems.
- Used Erwin to create relational conceptual, logical, and physical Data and Information Viewpoints (DIV 1, 2, and 3) and operational viewpoints graphics following DoDAF for FRCSs.
- Created notional data visualization dashboards comprehensively showcasing FRCS data in Army Vantage Big Data Platform.
- Conducted a gap analysis for various army suitable big data platforms.
- Created a normalized FRCS database and populated FRCS data tables in Microsoft SQL Server, eligible for robust analysis through SQL queries and Azure tools (Azure SQL Database, Data Factory etc).
- Displayed Azure tools such as Data Factory and Data Bricks as proof of concept that data can be easily accessible through Azure cloud.
- Illustrated proof of concept of uploading FRCS data files into Data Lake Gen2 Blob storage and an Extract, Transform, and Load (ETL) concept from Data Lake Gen2 into Azure SQL Database.
- Developed a threat and vulnerability identification guide for control system cybersecurity.

Enterprise Architect, ManTech (March 2021– May 2022)

Integrated Systems Architecture (ISA) project, Department of Homeland Security (DHS) / Customs Border Protection (CBP)

- Worked in the Strategic Capabilities Branch (SCB) under the Capabilities and Requirements Division (CRD) to support the Customs Border Protection (CPB) mission of identifying gaps in strategic initiatives.
- Used DoDAF principles and created architecture such various capability, operational, data/information and system viewpoints using Lifecycle Modeling Language (LML) and Unified Modeling Language (UML) for CBP programs using the software tool called Innoslate.
- Migrated from Innoslate to EKR ARIS diagramming tool.
- Created presentations and SharePoint pages on architectural content.
- Established metrics for conducting gap analysis on architecture.
- Updated high level and granular level CBP architecture with client approval.
- Contributed to System Engineering Lifecycle (SELC) and Acquisition Lifecycle Framework (ALF) methodologies within System Engineering Division (SED) team.
- Established an IT service management and engaged in a Dev Ops process for ISA team.
- Managed CBP architecture in database using Innoslate.
- Demonstrated proof of concept in Power BI showcasing data visualization functionality.
- Created a database in Microsoft SQL Server for CBP systems and migrated to Azure (used tools like Azure SQL Database, Data Factory, Data Bricks).
- Used Azure Data Factory to transform data stored in Data Lake Gen2 and loaded it into Azure SQL Database.

- Used Azure Databricks to store relational Data Warehouse of CBP system in which data was consistently added and created visualizations.
- Used ITIL concepts to establish service management after creating a website on the division's enterprise architecture branch.

Enterprise Data Modeler, Savvee Consulting (October 2019– March 2021)

Combined Business Architecture Requirements and Development (CBARD) project, Veteran Affairs (VA) / Veterans Health Administration (VHA)

- The primary area of focus was systems involved in supply chain management, where the aim was to eliminate redundant processes and assets in the strategic investment management branch.
- Used The Open Group Architecture Framework (TOGAF) principles of the Architecture Development Method (ADM) and modeled conceptual, logical, and physical UML class diagrams to visually depict business requirements, alongside subject matter expert (SME) feedback, using modeling tools of Lucid Charts, Rational System Architect (SA) and Cameo.
- Created information model requirement matrix's prior to creating "as is" and "to be" UML class data models, to organize and draft data into a comprehendible format.
- Developed a report called Information Model Summary Report (IMSR), which contributed to a comprehensive deliverable called Requirements Architecture Package (RAP).
- Migrated data models from RSA to Cameo during tool transition.
- Mentored and trained junior data modelers.
- Mapped data to business process models and business requirements.
- Maintained data model integrity by creating change logs.
- Used Power BI for data visualization and analytics.
- Engaged in medallion architecture, creating data models for bronze and silver layer databases in AWS Redshift using System Architect data modeling tool, for Supply Chain Master Database (SCMD) data.
- Explored non-relational database (NoSQL) options such as MongoDB and moon modeler for non-relational data models for supply chain data.

Business/Data Analyst, Nolij Consulting (November 2018 – October 2019)

Enterprise Architecture (EA) project, Department of Defense (DoD)/ Defense Health Agency (DHA)

- Assisted in the grand vision of the DHA of standardizing healthcare given to armed forces by providing data analytics for decision makers to make evidence-based decisions.
- Lead effort of consolidating all data. Created an EA data dictionary in Excel, consolidating different database data into one comprehensive source of information, using relational data modeling concepts.
- Worked in migrating EA data visualizations from a Resource Description Framework (RDF) format to H2 (Relational SQL format), using SQL joins and aggregations, and migrated database to SQL Server Management Studio (SSMS) for Azure related connections (Azure SQL Database, Data Factory etc).
- Manually created EA data visuals during SEMOSS upgrade.
- Lead an endeavor called Architectural Description Package (ADP), worked with enterprise architects to standardize DoDAF based architecture.
- Worked on endeavor called Architectural Maturity Inventory (AMI), updating enterprise architecture systems.
- Created a data visualization roadmap and management plan alongside subject matter experts (SMEs).
- Designed relational data models (conceptual, logical, and physical) in Erwin data modeling tool.
- Did data exchange analyses in SEMOSS for DoD healthcare systems.
- Used Python for creating Graphical User Interfaces (GUI's).
- Used Data Vault modeling techniques in Erwin.
- Briefed team on Microsoft Azure concepts in learning sessions (showed how SQL Server can connect to Azure SQL Database and be accessible with Azure Data Factory, Data Bricks, and how Blob features with Data Lake Gen2 can be used (Azure Data Factory for an ETL from Data Lake into Azure SQL Database, using data flows within Data Factory, Data Bricks for Data Warehousing capabilities, etc).

Data Analyst/Modeler, ITEC INC. (June 2014 – November 2018)

Trip Management System (TMS) project, NOAA (National Oceanic Atmospheric Administration)

- Worked through the entire software development life cycle alongside technical lead.

- Worked on creating a relational data model for TMS using Erwin to store trip notifications and universal trip id related information, a transactional database model to store detailed trip information.
- Familiar with dimensional data modeling, historical trip information was stored in a data warehouse.
- Used Tableau for data visualization to generate reports, data analytics and prediction.
- SQL components of Data Definition Language (DDL) and Data Manipulation Language (DML) were used in TMS project and tables were joined to analyze relationships using PostgreSQL.
- Used Python and PostgreSQL for ETL and used the trail feature of Azure to learn it for possible implementation.
- Used Tableau for data visualizations.